

Financial Output Check: Website vs Excel

Room-count comparison for the South/South-West Coast model using 50, 100, 150, 200 and 250 rooms. Website values use the current backend calculation engine. Excel values are reconstructed from the uploaded workbook formula structure and checked against the workbook dashboard sizing logic.

Main Reason For Difference

The mismatch is mainly assumption-driven. The current website default includes Sustainable Tourism Premium Market Scenario = Yes, laundry = Yes for this cluster, 2026 tariff basis, 25-year cash-flow period and 5.0% escalation. The workbook cash flow uses a 20-year basis, 2.5% escalation, biomass fuel as kWh_fuel cost, and 15-minute base-load export logic. Both sides must use the same switches and financial assumptions to match exactly.

Financial Outputs

Rooms	Metric	Website	Excel	Difference
50	NPV	LKR 551.03 M	LKR 207.55 M	LKR 343.48 M
50	IRR	36.73%	44.83%	-8.10%
50	Profitability index	3.39	3.24	0.15
50	Simple payback	3.16	2.19	0.97
50	Discounted payback	3.86	2.71	1.15
100	NPV	LKR 914.68 M	LKR 415.10 M	LKR 499.58 M
100	IRR	40.81%	44.83%	-4.02%
100	Profitability index	3.77	3.24	0.53
100	Simple payback	2.80	2.19	0.61
100	Discounted payback	3.37	2.71	0.66
150	NPV	LKR 1.36 B	LKR 551.02 M	LKR 806.21 M
150	IRR	42.51%	42.35%	0.16%
150	Profitability index	3.94	3.07	0.87
150	Simple payback	2.67	2.32	0.35
150	Discounted payback	3.20	2.89	0.31
200	NPV	LKR 1.72 B	LKR 758.57 M	LKR 962.31 M
200	IRR	43.88%	42.99%	0.89%
200	Profitability index	4.07	3.11	0.96
200	Simple payback	2.57	2.29	0.28
200	Discounted payback	3.07	2.84	0.23
250	NPV	LKR 2.22 B	LKR 894.49 M	LKR 1.32 B
250	IRR	43.03%	41.83%	1.20%
250	Profitability index	3.99	3.04	0.96
250	Simple payback	2.63	2.35	0.28
250	Discounted payback	3.15	2.93	0.22

Technical Cross-Check

Rooms	Turbine kW Web/Excel	CAPEX Web/Excel	Biomass t/y Web/Excel	Year 1 Net Cash Web/Excel
50	250 / 200	LKR 230.52 M / LKR 92.68 M	3131.1 / 2505.4	LKR -230.52 M / LKR 42.39 M
100	350 / 400	LKR 329.80 M / LKR 185.35 M	4383.6 / 5010.9	LKR -329.80 M / LKR 84.77 M
150	500 / 550	LKR 461.04 M / LKR 266.08 M	6262.3 / 6890.0	LKR -461.04 M / LKR 115.07 M
200	600 / 750	LKR 560.32 M / LKR 358.75 M	7514.7 / 9395.4	LKR -560.32 M / LKR 157.45 M
250	800 / 900	LKR 741.20 M / LKR 439.47 M	10019.6 / 11274.5	LKR -741.20 M / LKR 187.75 M

Conclusion

The corrected check uses workbook-style biomass fuel costing, turbine sizing and 15-minute export treatment. Remaining gaps are expected because the website scenario includes the sustainable tourism premium and different escalation/period assumptions. To force a like-for-like comparison, set the same sustainable-market option, analysis period, escalation rate, tariff year, laundry basis and CAPEX basis on both sides.